

AZ-900 Study Guide

AZ-900: Microsoft Azure Fundamentals — reference notes generated 2026-10-01. Use the official exam page for authoritative objectives.

About the exam

Audience

Business users, new IT staff, and anyone who needs foundational Azure and cloud vocabulary.

Format

The live exam is scored on a scale of up to 1000; ~700 is a common passing bar. This app uses 45 questions and a 70% practice threshold to mimic timing and breadth, not exact exam items.

Official objectives

<https://learn.microsoft.com/en-us/credentials/certifications/exams/az-900/>

This guide summarizes concepts only. Use the app's practice modes and Microsoft Learn modules for hands-on review.

Describe cloud concepts (25–30%)

What is cloud computing?

Cloud computing delivers compute, storage, networking, and applications over the internet with on-demand provisioning and pay-for-use pricing.

Key characteristics include broad network access, resource pooling, rapid elasticity, and measured service.

Cloud service models

- IaaS — You manage OS, middleware, runtime, data, and apps; provider manages physical infrastructure and virtualization.
- PaaS — You manage applications and data; provider manages runtime, middleware, OS, and infrastructure.
- SaaS — You consume software over the internet; provider manages nearly the entire stack.

Deployment models

- Public cloud — Shared provider infrastructure (e.g., Azure regions).
- Private cloud — Dedicated to one organization; may be on-premises or hosted.
- Hybrid cloud — Connects public and private with orchestration for migration and burst scenarios.

Economics and benefits

- CapEx vs OpEx — Cloud shifts many costs from upfront capital to operational spend.
- Elasticity and scalability — Scale out/in to match demand instead of over-provisioning hardware.
- High availability and fault tolerance — Design for redundancy across zones and regions.
- Consumption-based pricing — Meters usage; reservations and savings plans reduce cost for steady workloads.

Shared responsibility

Security and management duties depend on the service model. The provider always secures the physical datacenter and platform; the customer always owns data, identities, and access control choices.

SaaS shifts more responsibility to the provider; IaaS leaves more OS and network configuration to the customer.

Describe Azure architecture and services (35–40%)

Core architecture

- Regions and region pairs — Geographic areas for deployment; pairs support disaster recovery within a geography.
- Availability Zones — Physically separate datacenters within a region for resiliency.
- Resource groups — Logical containers for resources that share lifecycle.
- Subscriptions — Billing and access boundary; organized under management groups.
- Azure Resource Manager (ARM) — Deployment and management layer; templates enable infrastructure as code.

Compute

- Virtual Machines — Full control of the guest OS (IaaS).
- Virtual Machine Scale Sets — Identical VMs with autoscale.
- Azure App Service — Managed web and API hosting (PaaS).
- Azure Functions — Event-driven, serverless code.
- Azure Container Instances / AKS — Containers without or with Kubernetes orchestration.
- Azure Virtual Desktop — Desktop and app virtualization.

Networking

- Virtual Network (VNet) — Private network isolation in Azure.
- VPN Gateway and ExpressRoute — Hybrid connectivity (internet VPN vs private circuit).
- Azure DNS — Host DNS domains and records.
- Load Balancer and Application Gateway — Layer 4 vs Layer 7 traffic distribution; App Gateway can include WAF.
- Network Security Groups — Allow/deny traffic by IP, port, and protocol.

Storage

- Blob storage — Unstructured object data; tiers Hot, Cool, and Archive.
- Azure Files — SMB/NFS file shares.
- Disk Storage — Block storage for VMs.
- Table and Queue storage — NoSQL key-value and asynchronous messaging.
- Redundancy — LRS, ZRS, GRS, and geo-zone options trade cost vs durability.

Data and analytics (fundamentals level)

- Azure SQL Database — Managed relational PaaS.
- Azure Cosmos DB — Global, multi-model NoSQL with configurable consistency.

- Synapse Analytics — Enterprise analytics and data warehousing at a high level.

Identity and security services

- Microsoft Entra ID — Identity, SSO, and access for cloud applications.
- RBAC — Grant actions on scopes via built-in and custom roles.
- Conditional Access and MFA — Risk-based policies and multi-step sign-in.
- Microsoft Defender for Cloud — Secure score and workload protection recommendations.
- Key Vault — Central store for secrets, keys, and certificates.
- Azure Firewall, DDoS Protection, and Bastion — Network protection and secure admin access to VMs.

Describe Azure management and governance (30–35%)

Management tools

- Azure Portal — Web UI for resources.
- Azure Cloud Shell — Browser shell with Bash/PowerShell and CLI.
- Azure CLI and Azure PowerShell — Cross-platform and scriptable administration.
- ARM templates and Bicep — Repeatable, declarative deployments.
- Azure Arc — Extend Azure management to hybrid and multicloud resources.

Governance and compliance

- Management groups — Hierarchy above subscriptions for policy and RBAC inheritance.
- Azure Policy — Enforce organizational standards (allow/deny/remediate).
- Resource locks — CanNotDelete or ReadOnly protection.
- Tags — Metadata for cost allocation and automation.
- Microsoft Purview — Data governance and catalog (awareness level).
- Service Trust Portal — Compliance documentation and audit reports.

Monitoring and reliability

- Azure Monitor — Metrics, logs, alerts, and dashboards.
- Log Analytics — Query and analyze log data (KQL).
- Azure Service Health — Platform incidents and planned maintenance.
- Azure Advisor — Free recommendations for cost, security, reliability, performance, and operations.
- Well-Architected Framework — Five pillars: reliability, security, cost optimization, operational excellence, performance efficiency.

Cost management

- Azure pricing calculator — Estimate spend before deployment.
- Cost Management + Billing — Budgets, alerts, and analysis.
- TCO calculator — Compare on-premises vs Azure costs.
- Cost optimization — Right-size resources, reservations, and eliminating waste.

Microsoft Learn training map

Complete these free learning paths on Learn for coverage aligned with AZ-900 skill areas.

Introduction to Cloud Infrastructure: Describe cloud concepts

<https://learn.microsoft.com/en-us/training/paths/microsoft-azure-fundamentals-describe-cloud-concepts/>

Describe cloud computing — 24 min

<https://learn.microsoft.com/en-us/training/modules/describe-cloud-compute/>

Describe the benefits of using cloud services — 19 min

<https://learn.microsoft.com/en-us/training/modules/describe-benefits-use-cloud-services/>

Describe cloud service types — 13 min

<https://learn.microsoft.com/en-us/training/modules/describe-cloud-service-types/>

Introduction to Cloud Infrastructure: Describe Azure architecture and services

<https://learn.microsoft.com/en-us/training/paths/azure-fundamentals-describe-azure-architecture-services/>

Describe the core architectural components of Azure — 28 min

<https://learn.microsoft.com/en-us/training/modules/describe-core-architectural-components-of-azure/>

Describe Azure compute services — 33 min

<https://learn.microsoft.com/en-us/training/modules/describe-azure-compute-networking-services/>

Describe Azure networking services — 24 min

<https://learn.microsoft.com/en-us/training/modules/describe-azure-networking-services/>

Describe Azure storage services — 36 min

<https://learn.microsoft.com/en-us/training/modules/describe-azure-storage-services/>

Describe Azure identity, access, and security — 47 min

<https://learn.microsoft.com/en-us/training/modules/describe-azure-identity-access-security/>

Introduction to Cloud Infrastructure: Describe Azure management and governance

<https://learn.microsoft.com/en-us/training/paths/describe-azure-management-governance/>

Describe cost management in Azure — 27 min

<https://learn.microsoft.com/en-us/training/modules/describe-cost-management-azure/>

Describe features and tools in Azure for governance and compliance — 19 min

<https://learn.microsoft.com/en-us/training/modules/describe-features-tools-azure-for-governance-compliance/>

Describe features and tools for managing and deploying Azure resources — 22 min

<https://learn.microsoft.com/en-us/training/modules/describe-features-tools-manage-deploy-azure-resources/>

Describe monitoring tools in Azure — 13 min

<https://learn.microsoft.com/en-us/training/modules/describe-monitoring-tools-azure/>